

- **LLM Inference Engine**

A from-scratch LLM inference engine in PyTorch (KV cache, batching, paged memory)

[Github](#)

- Built a PyTorch LLM engine for GPT-2/Mistral with KV caching, continuous batching, and paged memory.
- Increased decode throughput 2.22x and cut CUDA time 46.6% with preallocated KV cache on A100.
- Scaled GPT-2 124M from 155 to 18,346 tokens/s (118.3x) at batch size 512 on A100.
- Cut peak KV memory 35.9%, reached 3.2x throughput at 64 concurrent requests, and added 122 tests.